Analysis of Two-Mode Networks Using R

2. Two-mode networks

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Outline

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Current version of slides (August 13, 2026 at 18 : 34): slides PDF
https://github.com/bavla/Nets/blob/master/ws/ws26.md

Two mode networks

A multipartite or multirelational network can be split into a collection of simpler one- or two-mode networks.

Often, such a collection is already produced in the phase of transforming source data into networks.

A typical example is a collection of bibliographic networks (Ci, WA, WK, WJ, WC, etc.) on a selected topic, created from data in bibliographic databases (Web of Science, Scopus, OpenAlex, etc.).

For creating bibliographic networks based on OpenAlex, we developed an R package `OpenAlex2Pajek`.

Two mode networks

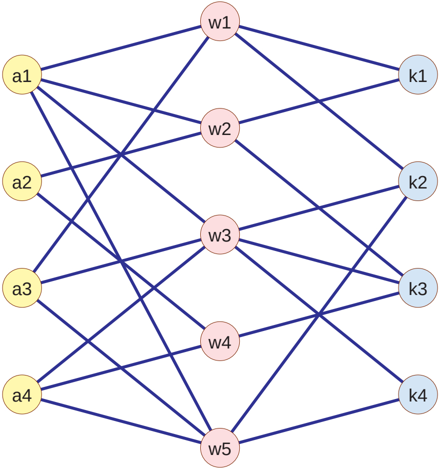
In igraph, a two-mode network has a logical node property `type` describing the partition of the set of nodes.

```
> library(jsonlite); library(httr)
> # AW <- read_graph("AW.net",format="pajek")
> AW <- read_graph(paste0(nWdir,"/data/AW.net"),format="pajek")
> AW
> V(AW)[[]]
> E(AW)[[]]
> is_bipartite(AW)
> AW <- delete_vertex_attr(AW,"type")
> AW
> is_bipartite(AW)
> V(AW)$type <- bipartite_mapping(AW)$type
> is_bipartite(AW)
> AW$name <- "Authors-Works"
> AW$date <- date()
> AW$twomode <- TRUE
> AW
> plot(AW,main=AW$name)

> # WK <- read_graph(WKfile,format="pajek")
> WK <- read_graph(paste0(nWdir,"/data/WK.net"),format="pajek")
> WK
```

Two mode networks

AW and WK – Authors, Works, and Keywords



Names of Participants of Group I	Code Numbers and Dates of Social Events Reported in *Old City Herald*													
	(1) 6/27	(2) 3/2	(3) 4/12	(4) 9/26	(5) 2/25	(6) 5/19	(7) 3/15	(8) 9/16	(9) 4/8	(10) 6/10	(11) 2/23	(12) 4/7	(13) 11/21	(14) 8/3
1. Mrs. Evelyn Jefferson	×	×	×	×	×	×		×	×					
2. Miss Laura Mandeville	×	×	×		×	×	×	×						
3. Miss Theresa Anderson		×	×	×	×	×	×	×	×					
4. Miss Brenda Rogers	×		×	×	×	×	×	×						
5. Miss Charlotte McDowd			×	×	×		×							
6. Miss Frances Anderson			×		×	×		×						
7. Miss Eleanor Nye					×	×	×	×						
8. Miss Pearl Oglethorpe						×		×	×					
9. Miss Ruth DeSand					×		×	×	×					
10. Miss Verne Sanderson							×	×	×			×		
11. Miss Myra Liddell								×	×	×		×		
12. Miss Katherine Rogers								×	×	×		×	×	×
13. Mrs. Sylvia Avondale							×	×	×	×		×	×	×
14. Mrs. Nora Fayette						×	×		×	×	×	×	×	×
15. Mrs. Helen Lloyd							×	×		×	×	×		
16. Mrs. Dorothy Murchison								×	×					
17. Mrs. Olivia Carleton									×		×			
18. Mrs. Flora Price									×		×			

A classical example of a two-mode network is the Davis Southern women's network [6].

```
> # davis <- read.csv(file.choose(), header=FALSE)
> davis <- read.csv(paste0(nWdir,"/data/davis.csv"))
> D <- graph_from_data_frame(davis, directed=FALSE)
> V(D)$type <- bipartite_mapping(D)$type
> V(D)$name[19:32] <- paste("event-",1:14,sep="")
> is_bipartite(D)
> D$name <- "Deep South"
> D$by <- "Davis, A., Gardner, B.B., Gardner, M.R."
> D$date <- 1941; D$twomode <- TRUE
> saveRDS(D,file="davis.rds")
```

Two mode network datasets

On GitHub/bavla, some examples of two-mode

Many additional two-mode networks can be found in indexes and repositories

- ICON - The Colorado Index of Complex Networks
- Netzschleuder
- Network Data Repository
- Kaggle
- UCI Network Data Repository
- KONECT - Koblenz Network Collection
- Bayesys
- Stanford Large Network Dataset Collection
- Pajek datasets

Named nodes!!!

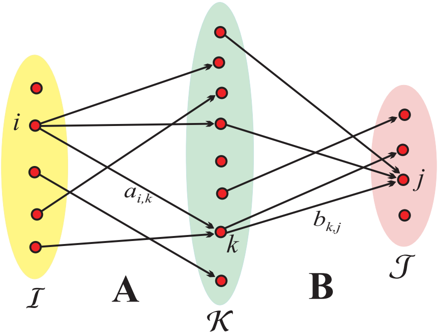
Multiplication of networks

Given a pair of compatible networks $\mathcal{N}_A = (\mathcal{I}, \mathcal{K}, \mathcal{A}_A, w_A)$ and $\mathcal{N}_B = (\mathcal{K}, \mathcal{J}, \mathcal{A}_B, w_B)$ with corresponding matrices $\mathbf{A}_{\mathcal{I}\times\mathcal{K}}$ and $\mathbf{B}_{\mathcal{K}\times\mathcal{J}}$ we call a *product of networks* $\mathcal{N}_A$ and $\mathcal{N}_B$ a network $\mathcal{N}_C = (\mathcal{I}, \mathcal{J}, \mathcal{A}_C, w_C)$, where $\mathcal{A}_C = \{(i,j) : i \in \mathcal{I}, j \in \mathcal{J}, c_{i,j} \neq 0\}$ and $w_C(i,j) = c_{i,j}$ for $(i,j) \in \mathcal{A}_C$. The product matrix $\mathbf{C} = [c_{i,j}]_{\mathcal{I}\times\mathcal{J}} = \mathbf{A} * \mathbf{B}$ is defined in the standard way

$$c_{i,j} = \sum_{k\in\mathcal{K}} a_{i,k} \cdot b_{k,j}$$

In the case when $\mathcal{I} = \mathcal{K} = \mathcal{J}$ we are dealing with ordinary one-mode networks (with square matrices).

Multiplication of networks



$$c_{i,j} = \sum_{k \in oN_A(i) \cap iN_B(j)} a_{i,k} \cdot b_{k,j}$$

If all weights in networks $\mathcal{N}_A$ and $\mathcal{N}_B$ are equal to 1 the value of $c_{i,j}$ counts the number of ways we can go from $i \in \mathcal{I}$ to $j \in \mathcal{J}$ passing through $\mathcal{K}$, $c_{i,j} = |oN_A(i) \cap iN_B(j)|$.

Multiplication of networks

The standard matrix multiplication has the complexity $O(|\mathcal{I}| \cdot |\mathcal{K}| \cdot |\mathcal{J}|)$ – it is too slow to be used for large networks. For sparse large networks we can multiply much faster considering only nonzero elements.

> **for** k **in** $\mathcal{K}$ **do**
>   **for** (i, j) **in** $iN_A(k) \times oN_B(k)$ **do**
>   **if** $\exists c_{i,j}$ **then** $c_{i,j} := c_{i,j} + a_{i,k} \cdot b_{k,j}$
>   **else** new $c_{i,j} := a_{i,k} \cdot b_{k,j}$

What is the meaning of the product network? In general, we could consider weights, addition and multiplication over a selected semiring [1]. In this presentation, we will limit our attention to the traditional addition and multiplication of real numbers.

In general, the multiplication of large, sparse networks is a 'dangerous' operation since the result can 'explode' – it is not sparse.

Multiplication of networks

From the network multiplication algorithm, we see that each intermediate node $k \in \mathcal{K}$ adds to a product network a complete two-mode subgraph $K_{iN_A(k),oN_B(k)}$ (or, in the case $\mathcal{I} = \mathcal{J}$, a complete subgraph $K_{oN(k)}$). If both degrees $\deg_A(k) = |iN_A(k)|$ and $\deg_B(k) = |N_B(k)|$ are large, then already the computation of this complete subgraph has a quadratic (time and space) complexity – the result 'explodes'.

If at least one of the sparse networks $\mathcal{N}_A$ and $\mathcal{N}_B$ has small maximal degree on $\mathcal{K}$ then also the resulting product network $\mathcal{N}_C$ is sparse.

If for the sparse networks $\mathcal{N}_A$ and $\mathcal{N}_B$ there are in $\mathcal{K}$ only few nodes with large degree and no one among them with large degree in both networks then also the resulting product network $\mathcal{N}_C$ is sparse.

Matrices

`igraph` provides a link to sparse matrices as supported in the package `Matrix`. When working with a single weight, it is usually useful to do computations with the corresponding sparse matrices obtained by the `netsWeight`'s function `as_sparse_matrix` and finally convert (if needed) the resulting matrix back to a network using `igraph`'s function `graph_from_matrix`.

*** Example

Derived networks

We can use multiplication to obtain new networks from existing *compatible* two-mode networks. For example, from basic bibliographic networks **WA** and **WK** we get

$$\mathbf{AK} = \mathbf{WA}^T \cdot \mathbf{WK}$$

a network relating authors to keywords used in their works, and

$$\mathbf{ACiA} = \mathbf{WA}^T \cdot \mathbf{Ci} \cdot \mathbf{WA}$$

is a network of citations between authors.
Networks obtained from existing networks using some operations are called *derived* networks. They are very important in the analysis of collections of *linked* networks.

The weight $\mathbf{AK}[a, k]$ is equal to the number of times the author a used the keyword k in his/her works.
The weight $\mathbf{ACiA}[a, b]$ counts the number of times a work authored by the author a is citing a work authored by the author b; or shorter, how many times the author a cited the author b?

...Derived networks

Using network multiplication, we can also transform a given two-mode network, for example **WA**, into corresponding ordinary one-mode networks (*projections*)

$$\mathrm{row}(\mathbf{WA}) = \mathbf{WW} = \mathbf{WA} \cdot \mathbf{WA}^T \quad \text{and} \quad \mathrm{col}(\mathbf{WA}) = \mathbf{AA} = \mathbf{WA}^T \cdot \mathbf{WA}$$

The obtained projections can be analyzed using standard network analysis methods. This is a traditional recipe for analyzing two-mode networks. Often the weights are not considered in the analysis; and when they are considered, we have to be very careful about their meaning.

The weight $\mathbf{WW}[p, q]$ is equal to the number of common authors of works p and q.
The weight $\mathbf{AA}[a, b]$ is equal to the number of works that authors a and b coauthored. In a special case, when $a = b$, it is equal to the number of works that the author a wrote. The network **AA** describes the *coauthorship* (collaboration) between authors and is also denoted as **Co** – the “first” coauthorship network.

Outer product decomposition

For vectors $x = [x_1, x_2, \ldots, x_n]$ and $y = [y_1, y_2, \ldots, y_m]$ their *outer product* $x \circ y$ is defined as a matrix

$$x \circ y = [x_i \cdot y_j]_{n \times m}$$

then we can express the previous observation about the structure of product network as the *outer product decomposition*

$$\mathbf{C} = \mathbf{A} \cdot \mathbf{B} = \sum_k \mathbf{H}_k \quad \text{where} \quad \mathbf{H}_k = \mathbf{A}[\cdot, k] \circ \mathbf{B}[k, \cdot]$$

For binary (weights) networks we have $\mathbf{H}_k = K_{oN_A(k), iN_B(k)}$.

Example A

Outer product decomposition

As an example let us take the binary network matrices **WA** and **WK**:

$$\mathbf{WA} = \begin{array}{c c} & \begin{array}{cccc} a_1 & a_2 & a_3 & a_4 \end{array} \\ \begin{array}{c} w_1 \\ w_2 \\ w_3 \\ w_4 \\ w_5 \end{array} & \begin{bmatrix} 1 & 0 & 1 & 0 \\ 1 & 1 & 0 & 0 \\ 1 & 0 & 1 & 1 \\ 0 & 1 & 0 & 1 \\ 1 & 0 & 1 & 1 \end{bmatrix} \end{array}, \quad \mathbf{WK} = \begin{array}{c c} & \begin{array}{cccc} k_1 & k_2 & k_3 & k_4 \end{array} \\ \begin{array}{c} w_1 \\ w_2 \\ w_3 \\ w_4 \\ w_5 \end{array} & \begin{bmatrix} 1 & 1 & 0 & 0 \\ 1 & 0 & 1 & 0 \\ 0 & 1 & 1 & 1 \\ 0 & 0 & 1 & 0 \\ 0 & 1 & 0 & 1 \end{bmatrix} \end{array}$$

and compute the product $\mathbf{H} = \mathbf{WA}^T \cdot \mathbf{WK}$. We get a network matrix **H** which can be decomposed as

. . . Example A

Outer product decomposition

$$\begin{array}{c|cccc} \mathbf{H} & k_1 & k_2 & k_3 & k_4 \\ \hline a_1 & 2 & 3 & 2 & 2 \\ a_2 & 1 & 0 & 2 & 0 \\ a_3 & 1 & 3 & 1 & 2 \\ a_4 & 0 & 2 & 2 & 2 \end{array} = \begin{array}{c|cccc} \mathbf{H}_1 & k_1 & k_2 & k_3 & k_4 \\ \hline a_1 & 1 & 1 & 0 & 0 \\ a_2 & 0 & 0 & 0 & 0 \\ a_3 & 1 & 1 & 0 & 0 \\ a_4 & 0 & 0 & 0 & 0 \end{array} + \begin{array}{c|cccc} \mathbf{H}_2 & k_1 & k_2 & k_3 & k_4 \\ \hline a_1 & 1 & 0 & 1 & 0 \\ a_2 & 1 & 0 & 1 & 0 \\ a_3 & 0 & 0 & 0 & 0 \\ a_4 & 0 & 0 & 0 & 0 \end{array} +$$

$$\begin{array}{c|cccc} \mathbf{H}_3 & k_1 & k_2 & k_3 & k_4 \\ \hline a_1 & 0 & 1 & 1 & 1 \\ a_2 & 0 & 0 & 0 & 0 \\ a_3 & 0 & 1 & 1 & 1 \\ a_4 & 0 & 1 & 1 & 1 \end{array} + \begin{array}{c|cccc} \mathbf{H}_4 & k_1 & k_2 & k_3 & k_4 \\ \hline a_1 & 0 & 0 & 0 & 0 \\ a_2 & 0 & 0 & 1 & 0 \\ a_3 & 0 & 0 & 0 & 0 \\ a_4 & 0 & 0 & 1 & 0 \end{array} + \begin{array}{c|cccc} \mathbf{H}_5 & k_1 & k_2 & k_3 & k_4 \\ \hline a_1 & 0 & 1 & 0 & 1 \\ a_2 & 0 & 0 & 0 & 0 \\ a_3 & 0 & 1 & 0 & 1 \\ a_4 & 0 & 1 & 0 & 1 \end{array}$$

In the paper [4] it was shown that there can be problems with the network **Co** when we try to use it for identifying the most collaborative authors. By the outer product decomposition, the coauthorship network **Co** is composed of complete subgraphs on the set of the work's coauthors.

. . . Outer product decomposition

Works with many authors produce large complete subgraphs, thus blurring the collaboration structure, and are over-represented by their total weight.

To see this, let $S_x = \sum_i x_i$ and $S_y = \sum_j y_j$ then the *contribution* of the outer product $x \circ y$ is equal

$$T_{xy} = \sum_{i,j} (x \circ y)_{ij} = \sum_i \sum_j x_i \cdot y_j = \sum_i x_i \cdot \sum_j y_j = S_x \cdot S_y$$

In general, each term $\mathbf{H}_w$ in the outer product decomposition of the product $\mathbf{C}$ has a different total weight $T(\mathbf{H}_w) = \sum_{a,k} (\mathbf{H}_w)_{ak}$, leading to over-representation of works with large values. In the case of coautorship network **Co** we have $S(\mathbf{WA}[w, .]) = \mathrm{odeg}_{\mathbf{WA}}(w)$ and therefore $T(\mathbf{H}_w) = \mathrm{odeg}_{\mathbf{WA}}(w)^2$. To resolve the problem, we apply the fractional approach.

Fractional approach

To make the contributions of all works equal, we can apply the *fractional* approach by normalizing the weights: setting $x' = x/S_x$ and $y' = y/S_y$, we get $S_{x'} = S_{y'} = 1$ and therefore $T(\mathbf{H}'_w) = 1$ for all works w.

In the case of two-mode networks **WA** and **WK** we denote

$$S_w^{\mathbf{WA}} = \begin{cases} \mathrm{wod}_{WA}(w) & \mathrm{odeg}_{\mathbf{WA}}(w) > 0 \\ 1 & \mathrm{odeg}_{\mathbf{WA}}(w) = 0 \end{cases} = \max(1, \mathrm{wod}_{WA}(w))$$

(and similarly $S_w^{\mathbf{WK}}$) and define the *normalized* matrices

$$\mathbf{WAn} = n(\mathbf{WA}) = \mathrm{diag}(\frac{1}{S_w^{\mathbf{WA}}}) \cdot \mathbf{WA}, \quad \mathbf{WKn} = n(\mathbf{WK}) = \mathrm{diag}(\frac{1}{S_w^{\mathbf{WK}}}) \cdot \mathbf{WK}$$

In real-life networks **WA** (or **WK**) some work may have no author. In such a case $S_w^{\mathbf{WA}} = \sum_a \mathbf{WA}[w, a] = 0$ which makes problems in the definition of the normalized network **WAn**. We can bypass the problem by setting $S_w^{\mathbf{WA}} = 1$, as we did in the above definition.

Normalized product

Then the *normalized product* matrix is $\mathbf{AKn} = \mathbf{WAn}^T \cdot \mathbf{WKn}$.

Denoting $\mathbf{F}_w = \frac{1}{S_w^{\mathbf{WA}} S_w^{\mathbf{WK}}} \mathbf{H}_w$ the outer product decomposition gets the form $\mathbf{AKn} = \sum_w \mathbf{F}_w$. Since

$$T(\mathbf{F}_w) = \begin{cases} 1 & (\mathrm{odeg}_{\mathbf{WA}}(w) > 0) \wedge (\mathrm{odeg}_{\mathbf{WK}}(w) > 0) \\ 0 & \text{otherwise} \end{cases}$$

we have further

$$\sum_{a,k} \mathbf{F}[a,k] = \sum_{a,k} \sum_w \mathbf{F}_w[a,k] = \sum_w T(\mathbf{F}_w) = |W^+|$$

where $W^+ = \{w \in W : (\mathrm{odeg}_{\mathbf{WA}}(w) > 0) \wedge (\mathrm{odeg}_{\mathbf{WK}}(w) > 0)\}$.

In the network **AKn**, the contribution of each work to the bibliography is 1. These contributions are redistributed to arcs from authors to keywords.

Example B

Outer product decomposition

For matrices from Example A, we get the corresponding diagonal normalization matrices

$$\mathrm{diag}\left(\frac{1}{S_w^{\mathbf{WA}}}\right) = \begin{array}{c} \\ w_1 \\ w_2 \\ w_3 \\ w_4 \\ w_5 \end{array} \begin{array}{c} \begin{array}{ccccc} w_1 & w_2 & w_3 & w_4 & w_5 \end{array} \\ \begin{bmatrix} 1/2 & 0 & 0 & 0 & 0 \\ 0 & 1/2 & 0 & 0 & 0 \\ 0 & 0 & 1/3 & 0 & 0 \\ 0 & 0 & 0 & 1/2 & 0 \\ 0 & 0 & 0 & 0 & 1/3 \end{bmatrix} \end{array}$$

$$\mathrm{diag}\left(\frac{1}{S_w^{\mathbf{WK}}}\right) = \begin{array}{c} \\ w_1 \\ w_2 \\ w_3 \\ w_4 \\ w_5 \end{array} \begin{array}{c} \begin{array}{ccccc} w_1 & w_2 & w_3 & w_4 & w_5 \end{array} \\ \begin{bmatrix} 1/2 & 0 & 0 & 0 & 0 \\ 0 & 1/2 & 0 & 0 & 0 \\ 0 & 0 & 1/3 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 1/2 \end{bmatrix} \end{array}$$

compute the normalized matrices

. . . Example B

Outer product decomposition

outer products such as

$$\mathbf{F}_1 = \begin{array}{c} \\ a_1 \\ a_2 \\ a_3 \\ a_4 \end{array} \begin{array}{c} \begin{array}{cccc} k_1 & k_2 & k_3 & k_4 \end{array} \\ \begin{bmatrix} 1/4 & 1/4 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 1/4 & 1/4 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix} \end{array} \qquad \mathbf{F}_5 = \begin{array}{c} \\ a_1 \\ a_2 \\ a_3 \\ a_4 \end{array} \begin{array}{c} \begin{array}{cccc} k_1 & k_2 & k_3 & k_4 \end{array} \\ \begin{bmatrix} 0 & 1/6 & 0 & 1/6 \\ 0 & 0 & 0 & 0 \\ 0 & 1/6 & 0 & 1/6 \\ 0 & 1/6 & 0 & 1/6 \end{bmatrix} \end{array}$$

and finally the product matrix $\mathbf{AKn} = \mathbf{WAn}^T \cdot \mathbf{WKn} =$

$$\sum_{w=1}^{5} \mathbf{F}_w = \begin{array}{c} \\ a_1 \\ a_2 \\ a_3 \\ a_4 \end{array} \begin{array}{c} \begin{array}{cccc} k_1 & k_2 & k_3 & k_4 \end{array} \\ \begin{bmatrix} 0.50000 & 0.52778 & 0.36111 & 0.27778 \\ 0.25000 & 0.00000 & 0.75000 & 0.00000 \\ 0.25000 & 0.52778 & 0.11111 & 0.27778 \\ 0.00000 & 0.27778 & 0.61111 & 0.27778 \end{bmatrix} \end{array}$$

Normalized co-authorship network

Let $\mathbf{N}$ be the normalized version $\forall w \in W : \sum_{a\in A} n[w,a] \in \{0,1\}$ obtained from $\mathbf{WA}$ by $n[w,a] = wa[w,a]/\max(1, \mathrm{odeg}_{WA}(w))$, or by some other rule determining the author's contribution – the *fractional* approach. Then the *normalized co-authorship network* is

$$\mathbf{Cn} = S(\mathbf{N}^T \cdot \mathbf{N})$$

$cn[a,b]$ = the total contribution of 'collaboration' of authors a and b to works.

It holds $\quad cn[a,b] = cn[b,a] \quad$ and for $w, \deg(w) > 0$ $\sum_{a\in A}\sum_{b\in A} n[w,a]\cdot n[w,b] = 1$.

The total contribution of a complete subgraph corresponding to the authors of a work p is 1. Therefore

$$\sum_{a\in A}\sum_{b\in A} cn[a,b] = |W^+|$$

$W^+ = \{w \in W : \deg(w) > 0\}$.

Normalized strict co-authorship network

Newman defined a *strict normalization* $\mathbf{N}'$ obtained from $\mathbf{WA}$ by $n'[w,a] = wa[w,a]/\max(1, \mathrm{odeg}_{WA}(w) - 1)$. Then the *normalized strict co-authorship network* is

$$\mathbf{Cs} = S(D_0(\mathbf{N}^T \cdot \mathbf{N}'))$$

Collaboration with **other** authors.

!!! The network **Cs** doesn't consider the contribution of single-author works.

Inner product

The *inner product* of vectors $x, y \in \mathbb{R}^n$ is defined as

$$\langle x, y \rangle = \sum_{i=1}^{n} x_i \cdot y_i$$

The following four properties hold for all $x, y, z \in \mathbb{R}^n$ and $\alpha \in \mathbb{R}$:

1. $\langle x, x \rangle \geq 0$ and $\langle x, x \rangle = 0$ if and only if $x = 0$,
2. $\langle x, y + z \rangle = \langle x, y \rangle + \langle x, z \rangle$,
3. $\langle x, \alpha y \rangle = \alpha \langle x, y \rangle$,
4. $\langle x, y \rangle = \langle y, x \rangle$.

An inner product $\langle ., . \rangle$ induces the *norm* of x

$$\|x\| = \sqrt{\langle x, x \rangle}$$

Cauchy-Schwarz inequality and Salton index

Cauchy-Schwarz inequality

$$| \langle x, y \rangle | \leq \|x\| \cdot \|y\|$$

Salton or cosine index

$$S(x, y) = \frac{\langle x, y \rangle}{\|x\| \cdot \|y\|} \in [-1, 1]$$

The Salton index has the following properties

1. $S(x, y) \in [-1, 1]$
2. $S(x, y) = S(y, x)$
3. $S(x, x) = 1$
4. $x, y \in (\mathbb{R}_0^+)^n \Rightarrow S(x, y) \in [0, 1]$
5. $S(\alpha x, \beta y) = S(x, y), \quad \alpha, \beta > 0$
6. $S(\alpha x, x) = 1, \quad \alpha > 0$

Salton index

Property 5 says that the value of the Salton index doesn't depend on the size of vectors but on their "shape". It is very useful when analyzing data with units of different "sizes" – for example, trade between world countries.

The Salton index is a measure of similarity. It is usually transformed into a dissimilarity d using the transformation

$$d(x, y) = \frac{1 - S(x, y)}{2}$$

or the angular distance θ wp

$$\theta(x, y) = \frac{\arccos(S(x, y))}{\pi}$$

Matrix based (dis)similarities

Let $\mathbf{W} = [w[u, v]]$ be a matrix describing (one/two-mode) (weighted) network $\mathcal{N}$. For clustering units (nodes) we need a dissimilarity matrix $\mathbf{D}$. For two-mode networks, we can get it from the Salton matrix $\mathbf{S} = [s[u, v]]$ where

$$s[u, v] = S(w[u, \cdot], w[v, \cdot])$$

If the matrix $\mathbf{W}$ is **one-mode** we have a problem

$$w[u, \cdot] = [w[u, 1], \ldots, w[u, z], \ldots, w[u, u], \ldots, w[u, v], \ldots, w[u, k]]$$

$$w[v, \cdot] = [w[v, 1], \ldots, w[v, z], \ldots, w[v, u], \ldots, w[v, v], \ldots, w[v, k]]$$

In traditional (dis)similarities, comparing $w[u, i]$ and $w[v, i]$ we are comparing how u relates to z with how v relates to z. The problem arises for $z = u$ and $z = v$. We would need to compare $w[u, u]$ with $w[v, v]$ and $w[u, v]$ with $w[v, u]$. This leads to *corrected* (dis)similarities.

Corrected Salton index

Corrected Salton index of the link $(u, v) \in \mathcal{L}$

$$S'(u, v) = \frac{\langle w[u, \cdot], w[v, \cdot] \rangle + (w[u, u] - w[u, v]) \cdot (w[v, v] - w[v, u])}{\|w[u, \cdot]\| \cdot \|w[v, \cdot]\|}$$

It preserves the usual Salton index properties 1-6.

Let $\mathcal{N} = ((\mathcal{U}, \mathcal{V}), \mathcal{L}, w)$ be a weighted two-mode network with a matrix $\mathbf{W} = [w[u, v]]$. For the matrix $\mathbf{Co} = [co[v, z]] = \mathbf{W}^T \cdot \mathbf{W}$ we have

$$co[v, z] = \sum_{u \in \mathcal{U}} w[v, u]^T \cdot w[u, z] = \sum_{u \in \mathcal{U}} w[u, v] \cdot w[u, z] = \langle w[v, \cdot], w[z, \cdot] \rangle$$

From which we can get the corresponding Salton matrix

$$s[u, v] = \frac{co[u, v]}{\sqrt{co[u, u] \cdot co[v, v]}}$$

Normalizations

in netsWeight

```
normalize_matrix_Markov(M)
normalize_matrix_Newman(M)
normalize_matrix_Balassa(M)
normalize_matrix_activity(M)
```

Normalizations

In networks obtained from large two-mode networks, there are often huge differences in weights. Therefore, it is not interesting to compare the nodes according to the raw data – the nodes/links with large values will prevail. First, we have to *normalize* the network to make the weights comparable.

... Normalizations

Markov (also known as stochastic, affinity, output, row) normalization: For $\mathrm{wod}(u) > 0$

$$w_M(u,v) = \frac{w(u,v)}{\mathrm{wod}(u)}$$

If $\mathrm{wod}(u) = 0$ also $w_M(u,v) = 0$.

Jaccard-like normalization:

$$w_J(u,v) = \frac{w(u,v)}{\mathrm{wod}(u) + \mathrm{wod}(v) - w(u,v)}$$

Salton-like normalization:

$$w_S(u,v) = \frac{w(u,v)}{\sqrt{\mathrm{wod}(u) \cdot \mathrm{wod}(v)}}$$

. . . Normalizations

Balassa or activity normalization [?]: Let $T = \sum_{u,v} w(u, v)$ be the total sum of weights in the network.

$$w_A(u, v) = \frac{w(u, v) \cdot T}{\text{wod}(u) \cdot \text{wid}(v)}$$

If $w_A(u, v) > 1$, the measured weight is larger than expected.

A symmetric version

$$w_B(u, v) = \log_2 w_A(u, v) \quad \text{for } w_A(u, v) > 0$$

... Normalizations

For a square matrix with nonzero diagonal values $\mathbf{C} = [c[u,v]]$ [2]

$$c_{\min}[u,v] = \frac{c[u,v]}{\min(c[u,u],c[v,v])} \qquad c_{\max}[u,v] = \frac{c[u,v]}{\max(c[u,u],c[v,v])}$$

$$c_{\mathrm{minDir}}[u,v] = \begin{cases} \dfrac{c[u,v]}{c[u,u])} & c[u,u] \leq c[v,v] \\ 0 & \textit{otherwise} \end{cases}$$

$$c_{\mathrm{maxDir}}[u,v] = \begin{cases} \dfrac{c[u,v]}{c[v,v]} & c[u,u] \leq c[v,v] \\ 0 & \textit{otherwise} \end{cases}$$

After a selected normalization, the important parts of the network are obtained by link-cuts or islands approaches.

Reuters Terror News: GeoDeg, MaxDir, MinDir. Slovenian journals.

Binarization and left and right (fractional) contribution

Binarization: $b(\mathbf{UV})$: $b(UV)[u,v] = \delta(uv[u,v] \neq \square)$

Left contribution: $L(\mathbf{UV}) = \mathbf{UV} \cdot b(\mathbf{UV})^T$

$$L(UV)[u,t] = \sum_{v \in V} uv[u,v] \cdot b(UV)[t,v] = \mathrm{wod}_{UV}(u/t)$$

Left fractional contribution: $\ell(\mathbf{UV}) = n(\mathbf{UV}) \cdot b(\mathbf{UV})^T$

$$\ell(UV)[u,t] = \frac{1}{\mathrm{wod}_{UV}(u)} \sum_{v \in V} uv[u,v] \cdot b(UV)[t,v] = \frac{\mathrm{wod}_{UV}(u/t)}{\mathrm{wod}_{UV}(u)} \leq 1$$

Right fractional contribution: $r(\mathbf{UV}) = b(\mathbf{UV}) \cdot n(\mathbf{UV})^T$

$$r(UV)[u,t] = \frac{1}{\mathrm{wod}_{UV}(t)} \sum_{v \in V} b(UV)[u,v] \cdot uv[t,v] = \frac{\mathrm{wod}_{UV}(t/u)}{\mathrm{wod}_{UV}(t)}$$

$$r(UV)[u,t] = \ell(UV)[t,u]$$

Mean value similarities

$$
\begin{aligned}
UU_X[u,t] &= meanX(\ell(UV)[u,t], r(UV)[u,t]) \\
&= meanX(\ell(UV)[u,t], \ell(UV)[t,u])
\end{aligned}
$$

$UU_A[u,t] = \frac{1}{2}(\ell(UV)[u,t] + \ell(UV)[t,u])$ – arithmetic mean

$UU_m[u,t] = \min(\ell(UV)[u,t], \ell(UV)[t,u])$ – minimum

$UU_M[u,t] = \max(\ell(UV)[u,t], \ell(UV)[t,u])$ – maximum

$UU_G[u,t] = \sqrt{\ell(UV)[u,t] \cdot \ell(UV)[t,u]}$ – geometric, Salton

$UU_H[u,t] = 2(\ell(UV)[u,t]^{-1} + \ell(UV)[t,u]^{-1})^{-1}$ – harmonic, Dice

$UU_J[u,t] = (\ell(UV)[u,t]^{-1} + \ell(UV)[t,u]^{-1} - 1)^{-1}$ – Jaccard

Note: $\ell(\mathbf{UV})$ can be computed from $L(\mathbf{UV})$
$L(UV)[u,u] = \mathrm{wod}_{UV}(u/u) = \mathrm{wod}_{UV}(u)$.

It holds: $UU_X[u,t] = UU_X[t,u]$, $UU_X[u,t] \in [0,1]$ and $UU_J[u,t] \leq$

Hamming distance

Co is a similarity measure. For clustering we need a dissimilarity.

A related dissimilarity between sets is the *Hamming distance*

$$d_H(A, B) = |A \oplus B| = |A \cup B \setminus A \cap B| = |A| + |B| - 2|A \cap B|$$

d_H is a distance

1. $d_H(A, B) = 0 \implies A = B$
2. $d_H(A, B) = d_H(B, A)$
3. $d_H(A, B) + d_H(B, C) \geq d_H(A, C)$

or in our case

$$D_H(v, z) = d_H(N_U(v), N_U(z)) = |N_U(v) \oplus N_U(z)|$$
$$= |Co[v, v]| + |Co[z, z]| - 2|Co[v, z]|$$

Note: $D_H(v, z) = 0$ implies $N_U(v) = N_U(z)$, but not $v = z$; v and z are structurally equivalent in the 2-mode network.

Binary similarity measures

The use of Jaccard similarity and some other similarities between binary vectors for analysis of two-mode networks was already proposed in SNA literature [**?**, p.420-424]. In principle, we could consider any similarity measure between binary vectors [**?**, **?**].

For all these similarities the corresponding matrices can be computed from the events co-affiliation or co-appearances matrix (ordinary column projection) **Co**. The similarities between vectors *v* and *z* are expressed in terms of the quantities *a*, *b*, *c*, and *d*

		z = 1	z = 0	
v	1	a	b	$a+b$
	0	c	d	$c+d$

Association coefficients

Association Coefficients

measure		definition	range	class
Russel and Rao (1940)	s_1	$\frac{a}{m}$	$[1,0]$	
Kendall, Sokal-Michener (1958)	s_2	$\frac{a+d}{m}$	$[1,0]$	S
Rogers and Tanimoto (1960)	s_3	$\frac{a+d}{m+b+c}$	$[1,0]$	S
Hamann (1961)	s_4	$\frac{a+d-b-c}{m}$	$[1,-1]$	S
Sokal & Sneath (1963), un_3^{-1}, S	s_5	$\frac{b+c}{a+d}$	$[0,\infty]$	S
Jaccard (1900)	s_6	$\frac{a}{a+b+c}$	$[1,0]$	T
Kulczynski (1927), T^{-1}	s_7	$\frac{a}{b+c}$	$[\infty,0]$	T
Dice (1945), Czekanowski (1913)	s_8	$\frac{a}{a+\frac{1}{2}(b+c)}$	$[1,0]$	T
Sokal and Sneath	s_9	$\frac{a}{a+2(b+c)}$	$[1,0]$	T
Kulczynski	s_{10}	$\frac{1}{2}(\frac{a}{a+b}+\frac{a}{a+c})$	$[1,0]$	
Sokal & Sneath (1963), un_4	s_{11}	$\frac{1}{4}(\frac{a}{a+b}+\frac{a}{a+c}+\frac{d}{d+b}+\frac{d}{d+c})$	$[1,0]$	
Q_0	s_{12}	$\frac{bc}{ad}$	$[0,\infty]$	Q
Yule (1912), ω	s_{13}	$\frac{\sqrt{ad}-\sqrt{bc}}{\sqrt{ad}+\sqrt{bc}}$	$[1,-1]$	Q
Yule (1927), Q	s_{14}	$\frac{ad-bc}{ad+bc}$	$[1,-1]$	Q
– bc –	s_{15}	$\frac{4bc}{m^2}$	$[0,1]$	
Driver & Kroeber (1932), Ochiai (1957)	s_{16}	$\frac{a}{\sqrt{(a+b)(a+c)}}$	$[1,0]$	
Sokal & Sneath (1963), un_5	s_{17}	$\frac{ad}{\sqrt{(a+b)(a+c)(d+b)(d+c)}}$	$[1,0]$	
Pearson, ϕ	s_{18}	$\frac{ad-bc}{\sqrt{(a+b)(a+c)(d+b)(d+c)}}$	$[1,-1]$	
Baroni-Urbani, Buser (1976), S^{**}	s_{19}	$\frac{a+\sqrt{ad}}{a+b+c+\sqrt{ad}}$	$[1,0]$	
Braun-Blanquet (1932)	s_{20}	$\frac{a}{\max(a+b,a+c)}$	$[1,0]$	
Simpson (1943)	s_{21}	$\frac{a}{\min(a+b,a+c)}$	$[1,0]$	
Michael (1920)	s_{22}	$\frac{4(ad-bc)}{(a+d)^2+(b+c)^2}$	$[1,-1]$	

Association coefficients

For example, the *Jaccard similarity*

$$J = \frac{|N_U(v) \cap N_U(z)|}{|N_U(v) \cup N_U(z)|} = \frac{a}{a+b+c}.$$

The following equalities hold

$$\begin{aligned} a &= |N_U(v) \cap N_U(z)| = Co[v,z] \\ a+b &= |N_U(v)| = \deg(v) = Co[v,v] \\ a+c &= |N_U(z)| = \deg(z) = Co[z,z] \\ a+b+c+d &= |U|. \end{aligned}$$

From them we get

$$\begin{aligned} b &= |N_U(v) \setminus N_U(z)| = Co[v,v] - Co[v,z] \\ c &= |N_U(z) \setminus N_U(v)| = Co[z,z] - Co[v,z] \\ d &= |U| + Co[v,z] - Co[v,v] - Co[z,z] \\ a+b+c &= |N_U(v) \cup N_U(z)| = Co[v,v] + Co[z,z] - Co[v,z] \end{aligned}$$

Jaccard similarity

For Jaccard similarity, we get [?]

$$Jcol[v, z] = \frac{Co[v, z]}{Co[v, v] + Co[z, z] - Co[v, z]}$$

In this sense, the similarity measures (matrices) can be seen as a kind of compatibility normalization of the weights obtained with the standard projection [?].

The corresponding Jaccard network $\mathcal{J} = (V, L_J, Jcol)$ is undirected with loops removed.

The Jaccard dissimilarity

$$d_J(A, B) = 1 - J(A, B) = \frac{|A \oplus B|}{|A \cup B|}$$

is also a distance. Note: $b + c = |N_U(v) \oplus N_U(z)|$.

Projections of two-mode networks

- **networks obtained by projection have a special structure – they are sums of complete subgraphs** [?].
- in creating the co-appearances network **Co** it is important to include the loops.
- the (a,b,c,d) (dis)similarities can be computed from the co-occurrence network **Co**.
- projections can be used to convert survey data (two-mode network) into a weighted network.

OpenAlex, client-libraries, OpenAlex2Pajek

```
> setwd(wdir <- "C:/Users/vlado/docs/papers/2025/AS/AMC")
> library(httr); library(jsonlite)
> source("https://raw.githubusercontent.com/bavla/Rnet/master/R/
> source("https://raw.githubusercontent.com/bavla/OpenAlex/
    main/code/OpenAlex2Pajek.R")
> sID <- "S61442588"
> R <- OpenAlexSources(sID,step=250)
OpenAlex2Pajek / Sources Mon May 26 05:59:58 2025
...
865 source S61442588 works collected Mon May 26 06:00:00 2025
..
 5231 citing works collected Mon May 26 06:05:46 2025
...
 12530 cited works collected Mon May 26 06:05:54 2025
12758 different works Mon May 26 06:05:54 2025
> csv <- file("worksAMC.csv","w",encoding="UTF-8")
> write(R,sep="\n",file=csv)
> close(csv)
```

```
> OpenAlex2PajekAll(NULL,name="AMC",listF="worksAMC.csv")
OpenAlex2Pajek / All - Start Mon May 26 06:10:05 2025
*** OpenAlex2Pajek / All - Process Mon May 26 06:10:05 2025
Mon May 26 06:13:38 2025  n = 500
Mon May 26 06:16:25 2025  n = 1000
...
Mon May 26 07:19:31 2025  n = 12500
*** OpenAlex2Pajek / All - Data Collected Mon May 26 07:20:44 20
hits: 12758 works: 137751 authors: 10849 anon: 185 sources: 1192
>>> Citation Cite
>>> publication year
>>> type of publication
>>> language of publication
>>> cited by count
>>> countries distinct count
>>> referenced works
>>> Authorship WA
>>> Sources WJ
>>> Countries WC
>>> Keywords WK
*** OpenAlex2Pajek / All - Stop Mon May 26 07:21:54 2025
```

```
> library(httr); library(jsonlite); library(pryr)
> OAdir <- "https://raw.githubusercontent.com/bavla/OpenAlex/"
> OAdata <- paste0(OAdir,"refs/heads/main/data/")
> OAcode <- paste0(OAdir,"main/code/")
> source(paste0(OAcode,"OpenAlex2Pajek.R"))

> WA <- read_graph(paste0(OAdata,"AMC/WA.net"),format="pajek")
> (nW <- sum(!V(WA)$type))
> (nA <- sum(V(WA)$type))

> (x <- exp(1))    # space needed
> bytes(x)
[1] "40 05 BF 0A 8B 14 57 69"
> as.integer(object.size(WA))/nW/nA/8
[1] 0.001895938

> WJ <- read_graph(paste0(OAdata,"AMC/WJ.net"),format="pajek")
> (nJ <- sum(V(WJ)$type))
> Ci <- read_graph(paste0(OAdata,"AMC/Ci.net"),format="pajek")

> source <- "S61442588"   # OA id of ACM
> (is <- which(V(WJ)$name==source))
[1] 137766
> s <- rep(0,nJ); s[is-nW] <- 1
> MWJ <- as_sparse_matrix(WJ)
> Wj <- MWJ %*% s
> sum(Wj)
[1] 865
```

We first clean the networks **Ci**, **WA**, **WJ**, ..., removing multiple links and loops.

The product $\mathbf{u} = \mathbf{A} \cdot \mathbf{v}$ of the network **A** with the vector **v** is defined as

$$u_i = \sum_{j:(i,j)\in L} A_{ij} \cdot v_j$$

We need the index j of the node $source =$ `"S61442588"` representing AMC in the set of journals J. We can get it from **WJ**: `is <- which(V(WJ)$name==source)` and $j = is - nW = 15$.

We start with the set W_j of all works published by the journal j.

$$W_j = \{w : WJ[w,j] > 0\}$$

Let $\mathbf{w}_j$ be its characteristic vector. Then $\mathbf{w}_j = \mathbf{WJ} \cdot [j]$ where $[j]$ is a vector over J having 1 at the jth place. We create the vector $[j]$

AMC analysis

works citing

Next, for the set W_j, we determine the set W_I of citing works and the set W_O of cited works.

$$W_I = \{w : \exists z \in W_j : Ci[w,z] > 0\} \quad \text{and} \quad W_O = \{w : \exists z \in W_j : Ci[z$$

The vectors $\mathbf{d}_I = \mathbf{Ci} \cdot \mathbf{w}_j$ and $\mathbf{d}_O = \mathbf{Ci}^T \cdot \mathbf{w}_j$

$$d_I(i) = \sum_k Ci[i,k] \cdot w_j(k), \quad d_O(i) = \sum_k Ci^T[i,k] \cdot w_j(k) = \sum_k Ci[k,i] \cdot w_j(k)$$

count: $d_I(i)$ - how many works from W_j are cited by the work i; and $d_O(i)$ - how many works from W_j are citing the work i.

Inspect the vector *dI*. We list the largest 20 nodes. We will collect from OpenAlex the additional information about the selected works.

It turns out that the authors' names are not directly accessible as a data field - they are contained inside the field "authorships". To extract them, we use the function `authors` embedded in the function `unitsInfo`.

Now we are ready to get the information about the selected works. Some data ("authors" and "title") can be very long. To get a readable report we truncate them.

```
> MCi <- as_sparse_matrix(Ci); namCi <- rownames(MCi)
> dI <- MCi %*% Wj
> p <- order(dI,decreasing=TRUE)
> selI <- p[1:20]
> (topI <- cbind(selI,namCi[selI],dI[selI]))
> selW <- paste0("id,language,countries_distinct_count,cited_by_
+   "relevance_score,publication_year,title,authorships")
> IWi <- unitsInfo(IDs=namCi[selI],units="works",select=selW,orc
> rep <- data.frame(id=IWi$id,cdc=IWi$countries_distinct_count,
+   cby=IWi$cited_by_count,dI=dI[selI],year=IWi$publication_year
+   authors=substr(authors(IWi),1,35),title=substr(IWi$title,1,4
> rep
```

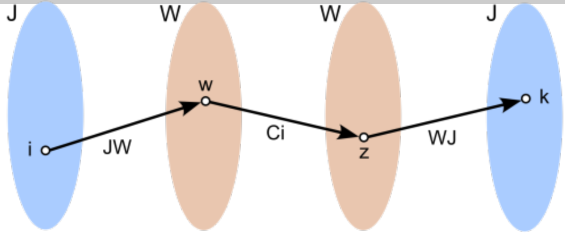
Some improvements: add source; in names, use only the last name, or initials + last name; ...

We can check selected works - for example W1846554597.

Using the same approach as for $\mathbf{d}_I$ we get also results for $\mathbf{d}_O$.

AMC analysis

Citations between journals



$$\mathbf{JJ} = \mathbf{WJ}^T \cdot \mathbf{Ci} \cdot \mathbf{WJ}$$

$JJ[i, k]$ = # of citations of a work from journal i to a work from journal k $\equiv$ # of times journal i cites journal k.

$n_{JJ} = 1192$, $m^A_{JJ} = 20011$, and 234 loops.

Inspecting the weights, we select the threshold $t = 100$. We make a link cut at level t.

AMC analysis

Citations between journals

```
> MJJ <- crossprod(MWJ,MCi) %*% MWJ
> nJJ <- nrow(MJJ); mJJ <- sum(MJJ>0); kJJ <- sum(diag(MJJ)>0)
> JJ <- graph_from_adjacency_matrix(MJJ,mode="directed",weighted
> JJ$project <- "Ars Mathematica Contemporanea"
> JJ$name <- "journal citation network"
> JJ$date <- date()
> JJ$by <- "Vladimir Batagelj"
> saveRDS(JJ,file="AMC_JJ.rds")

> w <- E(simplify(JJ))$weight
> r <- order(w,decreasing=TRUE)
> w[r[1:100]]
> LC <- link_cut(simplify(JJ),atn="weight",100)
> (S <- V(LC)$name)
> un <- 3; S[un] <- S[1]  # unknown
> selS <- "id,issn_l,country_code,type,is_oa,cited_by_count,
+   works_count,display_name"
> IS <- unitsInfo(IDs=S,units="sources",select=selS,order="input
> IS$display_name[un] <- IS$id[un] <- "Sunknown"; IS$issn_l[un]
> rep <- data.frame(id=IS$id,issn_l=IS$issn_l,journal=IS$display
> rep
> V(LC)$source <- IS$display_name
> lo <- layout_with_dh(LC)
> plot(LC,layout=lo,edge.width=log(w),vertex.color="pink",
+    vertex.size=12,vertex.label=V(LC)$source,vertex.label.cex=(
```

AMC analysis

Shorten the journal titles: Mathematics, Mathematical, Mathematische, Mathematicae $\rightarrow$ M; Combinatorics, Combinatorial $\rightarrow$ Comb; Computation, Computational $\rightarrow$ Comp; Computer Science $\rightarrow$ CS; Proceedings of the $\rightarrow$ P; Transactions of the $\rightarrow$ T; Bulletin of the $\rightarrow$ B; Journal of $\rightarrow$ J; Mathematical Society $\rightarrow$ MS; Discrete $\rightarrow$ Disc, etc.

Note, the network *WJ* is essentially a “function”.

There are other options to produce an ”interesting” subnetwork

- Extract the node cut for weighted degree at selected level.
- Determine the set of ”interesting” nodes as a weighted degree core (Ps-core).

AMC analysis

Citations between authors

$$\mathbf{ACiA} = \mathbf{WA}^T \cdot \mathbf{Ci} \cdot \mathbf{WA}$$

$ACiA[a, b]$ = # of citations of a work of author a to a work of author b ≡ # of times author a cites author b.

$n_{ACiA} = 10849$, $m^A_{ACiA} = 248183$, and 2251 loops.

Inspecting the weights we select the threshold $t = 100$. We make a link cut at level t.

To get a more readable network visualization, we have to replace IDs with the corresponding author names. The procedure is as in the case of journals. Normalized version!?

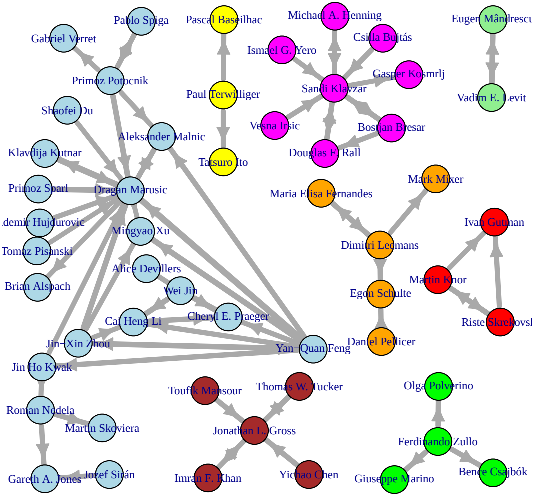
AMC authors

$$\mathbf{a}_j = \mathbf{WA}^T \cdot \mathbf{w}_j$$

$a_j(a)$ = # of works in the journal j co-authored by the author a

AMC analysis

Citations between authors



Conclusions

1. The package `netsWeight` is still in development. New functions will be added (islands, hubs and authorities, trusses, etc.) and it will probably merged with some other packages (`ClusNet`, `NetsJSON`,???)
2. The example is dealing with bibliographic network analysis. The proposed approach can be used for analysis of any collection of linked networks.

Acknowledgments

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References III

Two-mode networks 2

V. Batagelj

Two-mode networks

Multiplication

Fractional approach

Salton index

AMC

Conclusions

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